

The primal simplex method on a tableau

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1 Introduction

We consider a linear program (P) with n variables:

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax = b, \\ & x \geq 0_n, \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ is the coefficient matrix, $b \in \mathbb{R}^m$ denotes the right-hand-sides, and $c \in \mathbb{R}^n$ gives the objective coefficients. The set $N := \{1, \dots, n\}$ is used to index the variables. The columns of A are denoted A_j for $j \in N$.

1.1 Bases, basic solutions and reduced costs

The set $B \subseteq N$ will denote an arbitrary basis for (P), and $B = (i_1, \dots, i_m)$ denotes an ordering of the basic variables. The corresponding basis matrix is given by $A_B = [A_{i_1}, \dots, A_{i_m}]$. The set $J := N \setminus B$ is the index set for the non-basic variables. For every $j \in N$ let $\bar{A}_j := A_B^{-1} A_j$. The vector that contains only the cost coefficients of the basic variables is denoted:

$$c^B = \begin{pmatrix} c_{i_1} \\ \vdots \\ c_{i_m} \end{pmatrix}.$$

The basic solution associated with B is the unique solution $\bar{x} \in \mathbb{R}^n$ to the system $Ax = b$ and $x_j = 0$ for $j \in J$. We assume $\bar{x}$ is feasible for (P). A polyhedron $F_P(B)$ can be associated with the basis B :

$$F_P(B) := \{x \in \mathbb{R}^n : Ax = b \text{ and } x_j \geq 0 \text{ for } j \in J\},$$

and $F_P(B)$ is of the form $F_P(B) = \bar{x} + \text{Cone}(\{r^j\}_{j \in J})$, i.e., $F_P(B)$ is a translated cone. Every direction r^j with $j \in J$ is a unique solution to the system:

$$Ax = 0_m, \quad x_j = 1 \quad \text{and} \quad x_k = 0 \text{ for } k \in J \setminus \{j\},$$

and the coordinates of the direction r^j are given by the formula:

$$\begin{pmatrix} r_{i_1}^j \\ \vdots \\ r_{i_m}^j \end{pmatrix} = -\bar{A}_j, \quad \text{and for } k \in J : \quad r_k^j = \begin{cases} 1 & \text{if } k = j \\ 0 & \text{otherwise} \end{cases}$$

For every $j \in J$, the number $\bar{c}_j := c^T r^j$ is called the *reduced cost* associated with x_j . If $j \in J$ satisfies $\bar{c}_j < 0$, then all points on the halfline $H^j := \{\bar{x} + \alpha \cdot r^j : \alpha > 0\}$ have better objective value than $\bar{x}$. By construction all points on the halfline H^j satisfy the equalities $Ax = b$.

A formula for the reduced cost $\bar{c}_j$ associated with a non-basic variable $j \in J$ can be derived. One can split $c^T r^j$ into a basic part and a non-basic part, and then apply the formula for r^j :

$$\begin{aligned} \bar{c}_j = c^T r^j &= \sum_{k \in J} c_k r_k^j + \sum_{k \in B} c_k r_k^j \\ &= c_j + \sum_{k=1}^m c_{i_k} r_{i_k}^j = c_j - (c^B)^T \bar{A}_j \end{aligned}$$

2 Compact representation: The simplex tableau

We now derive a simple table that contains all the information needed for the primal simplex method. Every feasible solution to (LP) satisfies the equalities:

$$-Z + c^T x = 0 \tag{1}$$

$$Ax = b, \tag{2}$$

where Z is a variable that denotes the objective value of x . Multiplying equalities (2) with A_B^{-1} gives the following equalities that are equivalent to (1)-(2):

$$-Z + c^T x = 0 \tag{3}$$

$$\bar{A}x = \begin{pmatrix} \bar{x}_{i_1} \\ \vdots \\ \bar{x}_{i_m} \end{pmatrix} \tag{4}$$

Finally, multiplying both sides of (4) with $-c^B$ and adding the result to (3) gives:

$$-Z + (c - \bar{A}^T c^B)^T x = -c^T \bar{x} \tag{5}$$

$$\bar{A}x = \begin{pmatrix} \bar{x}_{i_1} \\ \vdots \\ \bar{x}_{i_m} \end{pmatrix} \tag{6}$$

Recall that the reduced costs of a non-basic variable $j \in J$ is given by $\bar{c}_j = c_j - (c^B)^T \bar{A}_j$. Therefore, defining the reduced costs to be zero for basic variables, *i.e.*, $\bar{c}_i = 0$ for $i \in B$, then (5) can be written $-Z + \bar{c}^T x = -c^T \bar{x}$. Figure 1 summerize equations (5)-(6). This table is called the *simplex tableau*.

Z	x_1	x_2	$\dots$	x_n	RHS
-1	$\bar{c}_1$	$\bar{c}_2$	$\dots$	$\bar{c}_n$	$-c^T \bar{x}$
0	$\vdots$	$\vdots$		$\vdots$	$\bar{x}_{i_1}$
$\vdots$	$\bar{A}_1$	$\bar{A}_2$	$\dots$	$\bar{A}_n$	$\vdots$
0	$\vdots$	$\vdots$		$\vdots$	$\bar{x}_{i_m}$

Figure 1 : The simplex tableau.

Observe that Z only appears in the first equality with a coefficient of (-1) . Also observe that, for every $k \in \{1, \dots, m\}$, the column of the simplex tableau corresponding to x_{i_k} is a unit vector. This holds since A_{i_k} is the k^{th} column of A_B , and therefore $\bar{A}_{i_k} = A_B^{-1} A_{i_k} = e_k$, where e_k is the k^{th} unit vector.

Hence the equalities contained in the simplex tableau are simply the equalities obtained by solving the system $Z = c^T x$ and $Ax = b$ wrt. Z and the basic variables $x_{i_1}, \dots, x_{i_m}$.

3 Example

We consider an example with $n = 4$ variables and $m = 2$ equality constraints. The coefficient matrix and the right-hand-side vector is given by:

$$A = \begin{pmatrix} -4 & 6 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 9 \\ 4 \end{pmatrix},$$

and the objective function coefficients are given by:

$$c^T = (-1 \quad 2 \quad 0 \quad 0).$$

The set $B = \{3, 4\}$ is a basis for (P) since the columns $A_3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $A_4 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ of A are linearly independent. We use the ordering $(i_1, i_2) = (3, 4)$, which gives the basis matrix $A_B = [A_3, A_4]$. Since A_B is the identity matrix, the basis inverse is $(A_B)^{-1} = A_B = I_2$. The basic solution $\bar{x}$ corresponding to B is:

$$\begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \bar{x}_3 \\ \bar{x}_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 9 \\ 4 \end{pmatrix},$$

and therefore $\bar{x}$ is a basic feasible solution. The objective value of $\bar{x}$ is $c^T \bar{x} = 0$. We will use the primal simplex method starting from this basis to solve the *maximization problem*. The ordering is $(i_1, i_2) = (3, 4)$, and therefore:

$$c^B = \begin{pmatrix} c_{i_1} \\ \vdots \\ c_{i_m} \end{pmatrix} = \begin{pmatrix} c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Since the basis matrix A_B is the identity matrix one has:

$$\bar{A}_1 = A_1 = \begin{pmatrix} -4 \\ 1 \end{pmatrix}, \bar{A}_2 = A_2 = \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \bar{A}_3 = A_3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } \bar{A}_4 = A_4 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Hence the reduced costs $\bar{c}_1$ and $\bar{c}_2$ for the non-basic variables are given by:

$$\bar{c}_1 = c_1 - (c^B)^T \bar{A}_1 = c_1 = -1 \quad \text{and} \quad \bar{c}_2 = c_2 - (c^B)^T \bar{A}_2 = c_2 = 2.$$

We now have all the data needed for the simplex tableau. The simplex tableau is shown in Figure 2.

Z	x_1	x_2	x_3	x_4	RHS
-1	-1	2	0	0	0 $(-c^T \bar{x})$
0	-4	6	1	0	9 $(\bar{x}_3)$
0	1	1	0	1	4 $(\bar{x}_4)$

Figure 2: The simplex tableau with x_3 and x_4 in the basis.

3.1 The first iteration of the primal simplex method

We next demonstrate how an iteration of the primal simplex method can be performed from the data given in the simplex tableau.

The first step of the primal simplex method is to identify a non-basic variable to enter the basis. This decision is based on the reduced costs $\bar{c}_j = c^T r^j$ of the non-basic variables $j \in J$. These numbers appear in the first row of the simplex tableau (see Figure 2). Since we are maximizing the objective, a non-basic variable $j_* \in J$ with a positive reduced cost is chosen.

For our example, the reduced costs on the non-basic variables x_1 and x_2 are $\bar{c}_1 = -1$ and $\bar{c}_2 = 2$. Hence we choose $j_* = 2$, *i.e.*, x_2 enters the basis.

The next step is to decide how large a value of $\theta \geq 0$ can be chosen such that $\bar{x} + \theta r^{j_*}$ is a feasible solution for (P), where r^{j_*} is the direction associated

with the non-basic variable x_{j_*} . The point $\bar{x} + \theta r^{j_*}$ is a feasible solution for (P) if and only if $\bar{x} + \theta r^{j_*} \geq 0_n$, and we have:

$$\begin{aligned} \bar{x} + \theta r^{j_*} \geq 0_n &\iff \theta \leq -\frac{\bar{x}_l}{r_l^{j_*}} \text{ for all } l \in \{1, \dots, n\} \text{ satisfying } r_l^{j_*} < 0 \\ &\iff \theta \leq \theta^* := \inf\left\{-\frac{\bar{x}_l}{r_l^{j_*}} : l \in \{1, \dots, n\} \text{ and } r_l^{j_*} < 0\right\}. \end{aligned}$$

If r^{j_*} does not have a negative coordinate, then (P) is unbounded, and the primal simplex method terminates.

Otherwise $\theta^* < +\infty$, and θ^* is obtained from the negative coordinates of r^{j_*} . The non-basic coordinates of r^{j_*} are all zero or one (see top of page 2), and therefore *not* negative. The basic coordinates of r^{j_*} are given by the formula:

$$\begin{pmatrix} r_{i_1}^{j_*} \\ \vdots \\ r_{i_m}^{j_*} \end{pmatrix} = -\bar{A}_{j_*} = -\begin{pmatrix} \bar{a}_{1,j_*} \\ \vdots \\ \bar{a}_{m,j_*} \end{pmatrix}.$$

It follows that we can give the following formula for θ^* :

$$\theta^* = \min\left\{\frac{\bar{x}_{i_k}}{\bar{a}_{k,j_*}} : k \in \{1, \dots, m\} \text{ and } \bar{a}_{k,j_*} > 0\right\}.$$

Hence θ^* can be computed from the basic coordinates of $\bar{x}$ and the positive coordinates of $\bar{A}_{j_*}$. This information is available in the j_* th column of the simplex tableau.

For our example we see from the simplex tableau in Figure 2 in the column corresponding to $x_{j_*} = x_2$ that $\bar{A}_{j_*} = \bar{A}_2 = \begin{pmatrix} 6 \\ 1 \end{pmatrix}$. We also see in the last column of the primal simplex tableau that the basic coordinates of the basic solution $\bar{x}$ are given by $\begin{pmatrix} \bar{x}_{i_1} \\ \bar{x}_{i_2} \end{pmatrix} = \begin{pmatrix} \bar{x}_3 \\ \bar{x}_4 \end{pmatrix} = \begin{pmatrix} 9 \\ 4 \end{pmatrix}$. We therefore obtain:

$$\theta^* = \min\left\{\frac{\bar{x}_{i_k}}{\bar{a}_{k,j_*}} : k \in \{1, 2\} \text{ and } \bar{a}_{k,j_*} > 0\right\} = \min\left\{\frac{9}{6}, \frac{4}{1}\right\} = \frac{3}{2}.$$

The next step is to find the leaving basic variable. The leaving basic variable is chosen among the basic variables that achieve the minimum above, *i.e.*, a basic variable i_{k_*} with $k_* \in \{1, \dots, m\}$ that satisfies $\bar{a}_{k_*,j_*} > 0$ and $\theta^* = \frac{\bar{x}_{i_{k_*}}}{\bar{a}_{k_*,j_*}}$.

In the example we have $j_* = 2$ and $(i_1, i_2) = (3, 4)$, and the minimum is $\theta^* = \frac{\bar{x}_{i_1}}{\bar{a}_{1,j_*}} = \frac{\bar{x}_3}{\bar{a}_{1,2}} = \frac{9}{6} = \frac{3}{2}$. Hence $x_{i_1} = x_3$ leaves the basis, so the new basis is:

$$B := (B \setminus \{3\}) \cup \{2\} = \{2, 4\}.$$

We have now demonstrated how to perform an iteration of the primal simplex method by using the data in the simplex tableau.

The next iteration starts with the basis $B = \{2, 4\}$. It is checked whether this basis is optimal, and if not, the basis is changed again. To do this one needs the simplex tableau associated with the basis $B = \{2, 4\}$. We now show how to obtain this (new) simplex tableau by performing row operations on the (old) simplex tableau in Figure 2.

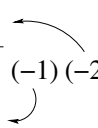
The simplex tableau in Figure 2 represents the equalities obtained by solving the system of equalities $Z = c^T x$ and $Ax = b$ wrt. the variable Z and the basic variables x_3 and x_4 . In order to obtain the simplex tableau associated with the basis $B = \{2, 4\}$, we need to solve the system $Z = c^T x$ and $Ax = b$ in terms of Z and the (new) basic variables x_2 and x_4 .

In terms of row operations on the simplex tableau, this corresponds to the following operations. First multiply the second row of the simplex tableau in Figure 2 with $\frac{1}{6}$:

Z	x_1	x_2	x_3	x_4	RHS
-1	-1	2	0	0	0
0	-4	6	1	0	$9 (\frac{1}{6})$
0	1	1	0	1	4

This results in the following table:

Z	x_1	x_2	x_3	x_4	RHS
-1	-1	2	0	0	0
0	$-\frac{2}{3}$	1	$\frac{1}{6}$	0	$\frac{3}{2}$
0	1	1	0	1	4



Next add multiples of the second row in the table above to the first row and the third row to make the column corresponding to x_2 a unit vector. This gives the following table:

Z	x_1	x_2	x_3	x_4	RHS
-1	$\frac{1}{3}$	0	$-\frac{1}{3}$	0	$-3(-c^T \bar{x})$
0	$-\frac{2}{3}$	1	$\frac{1}{6}$	0	$\frac{3}{2}(\bar{x}_2)$
0	$\frac{5}{3}$	0	$-\frac{1}{6}$	1	$\frac{5}{2}(\bar{x}_4)$

Figure 3: The simplex tableau with x_2 and x_4 in the basis.

The table in Figure 3 is the simplex tableau associated with the basis $B = \{2, 4\}$ and the basis matrix $A_B = [A_2, A_4]$. In other words, the table in Figure 3 gives the equalities:

$$\begin{pmatrix} -Z \\ x_2 \\ x_4 \end{pmatrix} + \begin{pmatrix} \frac{1}{3} \\ -\frac{3}{5} \\ \frac{1}{3} \end{pmatrix} x_1 + \begin{pmatrix} -\frac{1}{3} \\ \frac{1}{6} \\ -\frac{1}{6} \end{pmatrix} x_3 = \begin{pmatrix} -3 \\ \frac{3}{2} \\ \frac{5}{2} \end{pmatrix}.$$

Setting the non-basic variables x_1 and x_3 equal to zero in the above equality system gives the basic solution $\bar{x}$ associated with the basis $B = \{2, 4\}$:

$$\bar{x} = \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \bar{x}_3 \\ \bar{x}_4 \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{3}{2} \\ 0 \\ \frac{5}{2} \end{pmatrix}$$

Furthermore we see from the simplex tableau in Figure 3 that for the basis $B = \{2, 4\}$ we have:

$$\bar{A}_1 = \begin{pmatrix} -\frac{2}{3} \\ \frac{3}{5} \\ \frac{1}{3} \end{pmatrix}, \bar{A}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \bar{A}_3 = \begin{pmatrix} \frac{1}{6} \\ -\frac{1}{6} \end{pmatrix} \text{ and } \bar{A}_4 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Finally the first row of the simplex tableau in Figure 3 shows that the reduced costs associated with the basis $B = \{2, 4\}$ are given by:

$$\bar{c}_1 = \frac{1}{3}, \bar{c}_2 = 0, \bar{c}_3 = -\frac{1}{3} \text{ and } \bar{c}_4 = 0.$$

3.2 The remaining iterations on the simplex tableau

We now finish applying the primal simplex method on the example. We see from Figure 3 that the reduced cost on x_1 given by $\bar{c}_1 = \frac{1}{3}$ is positive. Since we are maximizing, this means it may be possible to find a better solution than $\bar{x}$ by increasing the value of x_1 . We therefore choose $x_{j_*} = x_1$ to enter the basis.

To find out which basic variable should leave the basis, we need to investigate how much we can increase the value of x_1 and remain feasible. First observe from the simplex tableau in Figure 3 that we have $\bar{A}_1 = \begin{pmatrix} -\frac{2}{3} \\ \frac{3}{5} \\ \frac{1}{3} \end{pmatrix}$. Hence the maximum amount by which we can increase x_1 and remain feasible is given by:

$$\theta^* = \min \left\{ \frac{\bar{x}_{i_k}}{\bar{a}_{k,j_*}} : k \in \{1, 2\} \text{ and } \bar{a}_{k,j_*} > 0 \right\} = \frac{\bar{x}_4}{\bar{a}_{2,1}} = \frac{5/2}{5/3} = \frac{3}{2}.$$

It follows that x_4 is the basic variable that must leave the basis since the minimum is achieved by $i_2 = 4$. The new basis is therefore:

$$B := (B \setminus \{4\}) \cup \{1\} = \{1, 2\}.$$

The next step is to create the simplex tableau associated with $B = \{1, 2\}$. This can be done with row operations: (a) First multiply the last row of the tableau in Figure 3 with $\frac{3}{5}$, then (b) multiply the last row in the table with $(-\frac{1}{3})$ and add the result to the first row, and finally (c) multiply the last row in the tableau with $(\frac{2}{3})$ and add the result to the second row:

Z	x_1	x_2	x_3	x_4	RHS
-1	$\frac{1}{3}$	0	$-\frac{1}{3}$	0	-3
0	$-\frac{2}{3}$	1	$\frac{1}{6}$	0	$\frac{3}{2}$
0	$\frac{5}{3}$	0	$-\frac{1}{6}$	1	$\frac{5}{2} (\frac{3}{5})$

Z	x_1	x_2	x_3	x_4	RHS
-1	$\frac{1}{3}$	0	$-\frac{1}{3}$	0	-3
0	$-\frac{2}{3}$	1	$\frac{1}{6}$	0	$\frac{3}{2}$
0	1	0	$-\frac{1}{10}$	$\frac{3}{5}$	$\frac{3}{2} (\frac{2}{3})(-\frac{1}{3})$

Z	x_1	x_2	x_3	x_4	RHS
-1	0	0	$-\frac{3}{10}$	$-\frac{1}{5}$	$-\frac{7}{2} (-c^T \bar{x})$
0	0	1	$\frac{1}{10}$	$\frac{2}{5}$	$\frac{5}{2} (\bar{x}_2)$
0	1	0	$-\frac{1}{10}$	$\frac{3}{5}$	$\frac{3}{2} (\bar{x}_1)$

Figure 4: Simplex tableau with x_1 and x_2 in the basis.

Figure 4 gives the simplex tableau associated with the basis $B = \{1, 2\}$. We see that the basic solution corresponding to $B = \{1, 2\}$ is given by;

$$\bar{x} = \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \bar{x}_3 \\ \bar{x}_4 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} \\ \frac{5}{2} \\ 0 \\ 0 \end{pmatrix}$$

with objective value $c^T \bar{x} = \frac{7}{2}$. Furthermore, the reduced costs associated with the basis $B = \{1, 2\}$ can be read from the first row of the final tableau:

$$\bar{c}_1 = 0, \bar{c}_2 = 0, \bar{c}_3 = -\frac{3}{10} \text{ and } \bar{c}_4 = -\frac{1}{5}.$$

Since we are maximizing, and all reduced costs are non-positive, we conclude that the basic solution $\bar{x}$ corresponding to the basis $B = \{1, 2\}$ defines an optimal solution to (LP).